Problem G
Fab Fury's Therapy Session

Time limit: 2 seconds

Fab Fury is a psychologist who helps her patients manage recurring thoughts and mental pressure through a series of therapy sessions.

She represents a patient's initial mental pressure by a positive integer $\boldsymbol{M}$.

During each session, Fab Fury chooses a prime number $\boldsymbol{p}$ that divides the patient's current mental pressure $\boldsymbol{x}$. The session costs

$$\boldsymbol{x} \times \mathbf{d}(\boldsymbol{p}),$$

where $\mathbf{d}(\boldsymbol{p})$ is the number of decimal digits in $\boldsymbol{p}$.

After the session, the patient's mental pressure is divided by $\boldsymbol{p}$, reducing it from $\boldsymbol{x}$ to $\boldsymbol{x/p}$. The cost is calculated using the value of $\boldsymbol{x}$ **before this division**. A prime may be chosen again in a later session if it still divides the current mental pressure.

Fab Fury continues conducting sessions until the patient's mental pressure becomes $\mathbf{1}$. Given the initial mental pressure $\boldsymbol{M}$, determine the minimum possible total cost of all sessions.

For example, when $\boldsymbol{M} = \mathbf{330}$, choosing the primes in the order $\mathbf{5, 3, 11, 2}$ gives a total cost of

$$\mathbf{330 + 66 + 44 + 2 = 442}.$$

However, choosing them in the order $\mathbf{5, 3, 2, 11}$ gives

$$\mathbf{330 + 66 + 22 + 22 = 440},$$

which is the minimum possible cost.

Input

The first line contains an integer $\boldsymbol{T}$ ($\mathbf{1} \leq \boldsymbol{T} \leq \mathbf{100000}$), the number of test cases.

Each of the next $\boldsymbol{T}$ lines contains one integer $\boldsymbol{M}$ ($\mathbf{1} \leq \boldsymbol{M} \leq \mathbf{10^{12}}$), the patient's initial mental pressure.

The sum of $\lfloor\sqrt{\boldsymbol{M}}\rfloor$ over all test cases is at most $\mathbf{2 \times 10^7}$.

Output

For each test case, print one line containing the minimum total cost of all sessions.

Sample Input 1	Sample Output 1
4 1 330 12 999983	0 440 18 5999898